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Original Article

Modeling epidemic and economic trade-offs under alternative COVID-19 isolation exit scenarios in South Korea

Suyoung Jo [a,b], Sung-il Cho [a,c], Asaph Young Chun [d,*,1], Kyung-Duk Min [b,*,1]

[a] Institute of Health and Environments, Seoul National University, Seoul, Republic of Korea
[b] College of Veterinary Medicine, Chungbuk National University, Cheongju, Republic of Korea
[c] Graduate School of Public Health, Seoul National University, Seoul, Republic of Korea
[d] Institute for Pandemic Sciences AI.celerator, Seoul National University, Seoul, Republic of Korea

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ABSTRACT

Background: Scientific evidence should guide adjustments to isolation policies as part of coronavirus disease 2019 (COVID-19) exit strategies. However, such evidence was limited when these strategies were implemented. This study retrospectively evaluated the relaxation of isolation policies in South Korea, aiming to generate evidence to support policy decisions in future pandemic scenarios.
Methods: We developed an age-stratified deterministic vaccination-extended Susceptible–Exposed–Infectious–Recovered model to simulate counterfactual scenarios related to isolation policy changes during the Omicron wave. Two phases were assessed: partial relaxation of isolation in 2022 and complete lifting of isolation mandates in 2023. Model outputs included confirmed cases and wage loss. Modified cost-effectiveness ratios were calculated as economic savings per additional infection.
Results: Moderate shortening of isolation from 10 to 7 days before the epidemic peak was associated with relatively favorable trade-offs between additional infections and wage loss reduction. In contrast, a more aggressive approach, reducing isolation to 5 days, resulted in a substantially higher number of confirmed cases with only marginal additional economic benefit. Delaying full lifting until July 2023, 1 month after the actual policy change, led to smaller and later epidemic peaks and greater economic gains.
Conclusions: Our findings suggested that both the timing and extent of isolation policy relaxation significantly influence epidemic dynamics and economic-epidemiologic trade-offs. Scenario-based modeling offers a valuable tool for anticipating trade-offs and informing policy decisions. Strengthening real-time surveillance and predictive modeling infrastructure is critical for an adaptive and balanced pandemic response.
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Introduction

Isolation of confirmed coronavirus disease 2019 (COVID-19) cases has been a cornerstone of such interventions [1,2], effectively reducing secondary transmission by removing infectious individuals from the community. Throughout most of the pandemic, isolation was legally mandated and enforced in South Korea, with penalties for non-compliance. During the earlier phases of the pandemic, case identification relied exclusively on reverse transcription polymerase chain reaction (RT-PCR) testing within the national surveillance system, and isolated individuals were actively monitored by local public health authorities. As the Omicron variant led to a rapid surge in cases, case confirmation was expanded to include clinician-administered rapid antigen tests (RAT), and the isolation system shifted toward a predominantly home-based self-isolation system for the general population, while high-risk individuals continued to be managed in clinical or institutional settings [3]. While isolation has proven effective in suppressing viral spread, mandatory isolation also entails broader societal and socioeconomic costs [4–6]. In this context, the mandatory isolation period for confirmed cases was reduced from 10 to 7 days in early 2022 and was completely lifted by mid-2023 as part of the South Korea's pandemic exit strategy [7].

Abbreviations: COVID-19, Coronavirus Disease 2019; KDCA, Korea Disease Control and Prevention Agency; ODE, Ordinary Differential Equation; RAT, Rapid Antigen Tests; RT-PCR, Reverse Transcription Polymerase Chain Reaction

* Corresponding authors.
E-mail addresses: ychun2@gmail.com (A.Y. Chun), kdmin@chungbuk.ac.kr (K.-D. Min).
[1] These authors contributed equally as corresponding authors

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These policy decisions occurred amidst evolving scientific knowledge, increasing population-level immunity [8], shifting public risk perceptions [9], and growing economic concerns [10] during the Omicron wave. Given the urgency and uncertainty of the evolving situation, many policy shifts were implemented without comprehensive evaluations of their potential trade-offs.

During the COVID-19 pandemic, a range of modeling approaches, including time-series analysis [11,12], machine learning [13,14], and mathematical modeling [15,16], were used to evaluate public health measures for infectious disease control. Among these, scenario-based mathematical transmission models are particularly well-suited for assessing the impact of policy interventions under uncertainty [17]. Such models allow for comparisons between observed and counterfactual scenarios and can jointly estimate health and economic outcomes within a unified framework [18]. This enables the assessment of trade-offs between competing priorities, such as reducing infections versus minimizing productivity loss—trade-offs that retrospective observational analyses or single-outcome evaluations cannot fully capture. While scenario-based models have been widely used to evaluate containment strategies, such as vaccination and social distancing [15,16,18], evaluations of relaxed isolation policies remain limited, particularly during the later phases of the pandemic, when the policy objective shifted from elimination to the sustained management of transmission.

Moreover, most evaluations of COVID-19 interventions have focused primarily on epidemiological outcomes, such as confirmed cases, hospitalizations, and deaths. Although these metrics are critical, they are insufficient for informing balanced policy decisions. Public health measures, such as isolation mandates, also generate broader societal burdens, including disruptions to economic activity resulting from the temporary removal of individuals from the workforce [19]. However, most economic evaluations have primarily focused on direct medical costs, whereas broader societal costs beyond healthcare expenditures have received limited attention, particularly in the context of isolation policies. Among these, productivity loss—reflecting workdays lost due to mandatory isolation—constitutes a key component of the societal burden associated with isolation policies. This highlights a critical gap in understanding of the full societal impact of such interventions [20,21]. In this context, mathematical modeling offers a valuable framework for assessing both health and economic outcomes, providing insights into the trade-offs that policymakers must consider when adjusting interventions like isolation mandates.

South Korea presents a particularly suitable setting for this analysis, as its isolation policies were both legally mandated and widely followed by the public. Against this background, we developed a mathematical model calibrated to South Korea's national data to evaluate the epidemiological and productivity-related consequences of relaxing isolation mandates. Our analysis focused on two key policy shifts: the partial shortening of the isolation period in early 2022 and its complete removal in mid-2023. By simulating counterfactual scenarios that vary in timing and intensity, we aimed to provide insights into how adjustments to isolation policies may influence infection trends and productivity loss. Ultimately, our goal was to examine how the timing and extent of isolation policy changes shape the trade-offs between epidemic control and productivity-related societal burden.

Material and methods

Data source

This study utilized publicly available data from two governmental agencies: the Korea Disease Control and Prevention Agency (KDCA) and Statistics Korea. The KDCA provided age-specific data on daily new confirmed COVID-19 cases, defined as individuals with positive RT-PCR test results. Additionally, age-stratified information was collected on the daily number of individuals receiving their primary COVID-19 vaccinations and booster shots. Population data for 2021 were obtained from Statistics Korea. All aggregated data used in this study were collected over the period from 16 January 2022–31 August 2023.

To inform the economic evaluation, national statistics on average daily wages, labor force participation rates, and monthly exchange rates were obtained from e-Nara Indicators, an official statistical portal operated by the Korean government (https://www.index.go.kr). These indicators were used to calculate productivity losses associated with isolation based on age-specific employment and income characteristics during the study period.

Model structure and parameterization

A deterministic, vaccination-extended Susceptible–Exposed–Infectious–Recovered compartmental model was developed to simulate COVID-19 transmission dynamics and to evaluate the impact of lifting isolation measures during the pandemic in South Korea. The study period spanned from 16 January 2022–31 August 2023, covering the Omicron-dominant phase. To reflect major shifts in public health policy and epidemic intensity, the period was divided into two phases: Phase 1 (16 January to 31 December 2022) and Phase 2 (1 January to 31 August 2023).

The model incorporated the natural history of COVID-19, trends in vaccination coverage, and changes in isolation policies specific to South Korea. The population was stratified into three age groups (0–17, 18–59, and ≥ 60 years), with individuals progressing through the compartments of Susceptible (S), Vaccinated (V), Exposed (E), Infectious (I), and Removed (R) (Fig. 1). The vaccinated compartment was further divided into three stages, V_1 (first dose), V_2 (second dose), and V_3 (booster), based on vaccination history. Vaccination counts (v_1, v_2, v_3) were obtained from the national KDCA database. Individuals receiving their first, second, and booster doses sequentially as they transitioned from V_1 to V_2 to V_3. Dose-specific vaccine effectiveness was set at 42.8 %, 65.5 %, and 67.2 % for V_1, V_2, and V_3, respectively (ε_1, ε_2, ε_3) [22]. Vaccine-induced immunity was assumed to wane after an average of 138 days (ω_1^{-1}) [23], after which individuals returned to the susceptible compartment. Initial values for all model compartments at the start of the simulation period were specified by age group and are reported in Supplementary Material 2.

Susceptible individuals were exposed to infectious individuals at a force of infection (λ), defined as the product of the effective contact rate (β) and the proportion of infectious individuals in the population. The effective contact rate was modeled as the product of the contact rate and the probability of transmission per contact. To account for temporal and age-specific variations, β was estimated weekly by age group and calibrated using a local random search algorithm to fit the observed daily number of confirmed cases.

Exposed individuals (E) transitioned to the infectious compartment (I) after a latent period, assumed to be 1.5 days (θ^{-1}), based on published estimates of the incubation period for the Omicron variant [24]. Infectious individuals in compartment I were then differentiated according to detection and isolation status. A proportion μ of infectious individuals who were confirmed and placed under isolation, moved to the R_1 compartment, corresponding to the daily number of confirmed cases reported by the KDCA. The remaining proportion ($1-\mu$) represented undetected infectious individuals who entered the I_u compartment and subsequently progressed to R_u while retaining their infectiousness.

The full isolation period (δ^{-1}) was assumed to be 10 days [25]. As isolation policies were relaxed, individuals who remained infectious upon completing isolation transitioned to the I_2 compartment before moving to R_2, representing those who had fully recovered and were

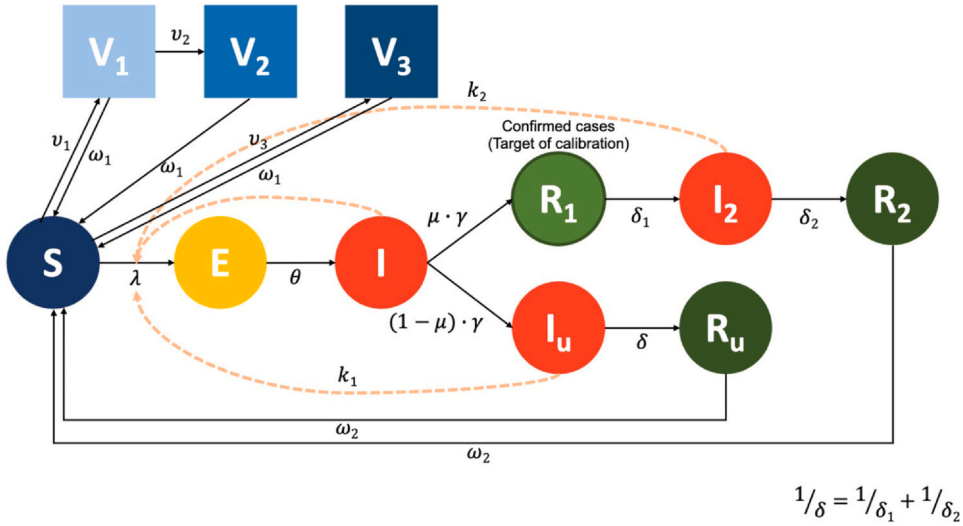

Fig. 1. Schematic diagram of the mathematical model. Solid black arrows represent transitions between compartments, while dashed orange arrows indicate transmission pathways from infectious to susceptible individuals. The system of ordinary differential equations (ODEs) governing the model is detailed in Supplementary Material 1.

no longer infectious. In cases where the isolation duration was shortened, partially infectious individuals exited isolation early and entered the I_2 compartment.

Accordingly, the force of infection incorporated contributions from all infectious states, including detected infectious individuals (I), undetected infectious individuals (I_u), and partially infectious individuals following shortened isolation (I_2). To account for reduced infectivity in these compartments, relative infectiousness adjustment factors (k_1 and k_2) were applied based on a previous study [26]. The detection period (γ^{-1}) was fixed at 5.13 days [27], while μ was estimated through model calibration. Infection-derived immunity was assumed to wane after 480 days ($\omega_2{}^{-1}$) [28], after which the individuals returned to the susceptible state.

Table 1 summarizes the key model parameters, including fixed values, data-driven estimates, and calibration targets.

Counterfactual scenarios for evaluation

To evaluate the impact of lifting isolation policies, the Omicron wave was divided into two distinct periods, each marked by major changes in national isolation protocols. Table 2 presents the scenario structure for each period.

Period 1: 2022—Reduction in isolation duration

On 10 February 2022, the mandatory isolation period for confirmed COVID-19 cases was shortened from 10 days to 7 days. To assess the potential epidemiological and economic consequences of this policy change, we designed counterfactual scenarios that varied both the timing and extent of the reduction in isolation duration. Specifically, we evaluated a scenario in which implementation of the policy was delayed until 22 April 2022, the point at which the model-estimated proportion of immune individuals in the economically active age group (18–59 years) reached its peak (Supplementary Material 3).

Table 2
Scenarios for evaluation of COVID-19 isolation policies in South Korea.

Policy Period: 16 January 2022–31 December 2022		
Scenario	**Description**	**Date of implementation**
#1	Maintain 10-day isolation requirement	N/A (baseline)
#2*	Reduce isolation period to 7 days	10 February 2022
#3	Reduce isolation period to 5 days	10 February 2022
#4	Reduce isolation period to 7 days	22 April 2022[a]
#5	Reduce isolation period to 5 days	22 April 2022[a]
Policy Period: January 16, 2022–December 31, 2022		
Scenario	**Description**	**Date of implementation**
#1	Lift mandatory isolation	1 February 2023
#2	Lift mandatory isolation	1 March 2023
#3	Lift mandatory isolation	1 April 2023
#4	Lift mandatory isolation	1 May 2023
#5*	Lift mandatory isolation	1 June 2023
#6	Lift mandatory isolation	1 July 2023
#7	Lift mandatory isolation	1 August 2023 (baseline)

Baseline scenarios represent the most stringent isolation policy in each period.
* Policy actually implemented
[a] Immunity level in the 18–59 years age group peaked on 22 April 2023

Table 1
Model parameters and assumptions in this model in the age-structured COVID-19 isolation policy model.

Category	Parameter	Value
Transmission related	Effective contact rate (β)	Calibrated
	Latent period of omicron variant (θ^{-1})	3.5 − 2 days [24]
	Detection period (γ^{-1})	3.13 + 2 days [27]
	Probability of detection (μ)	Calibrated
	Full isolation duration (δ^{-1})	10 days [25]
	Relative ratio of infectivity (k)	Estimated from ref [26]
Immunization related	Immunization period of vaccination (ω_1)	138 days [23]
	Immunization period of infection (ω_2)	480 days [28]
	Daily vaccine administration rate (v_1, v_2, v_3)	Observed data
	Effectiveness of the first dose of the primary vaccination series (ε_1)	0.428 [22]
	Effectiveness of the second dose of the primary vaccination series (ε_2)	0.655 [22]
	Effectiveness of booster shot (ε_3)	0.672 [22]

Period 2: 2023—Full REmoval of Mandatory Isolation

On 1 June 2023, the mandatory isolation requirement for confirmed cases was lifted entirely. Scenarios for this period were designed to assess the optimal timing for this policy change. We simulated alternative dates for lifting the isolation mandate and compared outcomes to determine whether a different implementation date might have achieved more favorable epidemiological and economic results under comparable conditions.

Economic evaluation of isolation policy scenarios

To assess the economic efficiency of each isolation policy scenario, we calculated a modified cost-effectiveness ratio, defined as the reduction in total wage loss per additional confirmed case relative to a baseline scenario. For each analytic period (2022 and 2023), the baseline scenario was defined as the one with the most stringent isolation requirement, serving as the reference for comparison. Total wage loss was estimated based on confirmed cases among the economically active population aged 18–59 years, under the assumption that individuals were unable to work during their isolation period. Similar wage-based valuation has been used in prior studies estimating productivity losses due to COVID-19-related work absenteeism under a human capital framework [29,30].

According to national statistics, the labor force participation rate in this age group was 63.9 % in 2022 and 64.3 % in 2023. For each confirmed case, we assumed a loss of five workdays (accounting for a 7-day isolation period with two weekend days excluded), which was then multiplied by the average daily wage: 197,398 KRW in 2022 and 204,433 KRW in 2023. Wage loss estimates were converted to US dollars using the average monthly exchange rate at the time of infection.

Unlike conventional cost-effectiveness analyses, which estimate additional costs per unit of health gain, we adapted the concept in this study as a modified cost-effectiveness ratio, quantifying the economic benefit, measured as reduced wage loss, per additional confirmed case resulting from a less stringent isolation policy. This approach provides a policy-relevant metric for evaluating the trade-off between increased transmission and earlier socioeconomic recovery.

Statistical analysis

All simulations and analyses were conducted using R software (version 4.2.1; R Foundation for Statistical Computing, Vienna, Austria). Scenario-based deterministic simulations were conducted under alternative isolation policy assumptions. Accordingly, the analyses were descriptive and comparative in nature rather than inferential, and no formal statistical hypothesis testing was applied.

Results

Impact of reducing the isolation period (Period 1: January–December 2022)

The simulation results for Period 1 (January to December 2022) are presented in Fig. 2, which evaluates the impact of reducing the mandatory isolation period from 10 days and the timing of this reduction. Panel (A) shows the model fit to observed confirmed case data across age groups, while Panel (B) presents the projected epidemic trajectories under five different scenarios, including the actual policy implemented in February 2022. The model adequately captured the timing and relative magnitude of epidemic peaks by age group. While age-specific transmission patterns remained generally consistent across scenarios, differences became more pronounced toward the end of 2022, particularly among younger individuals.

The timing of the policy change had a relatively limited effect on overall transmission; however, the extent of isolation reduction (i.e., from 10 to 7 or 5 days) had a more substantial impact on the cumulative number of confirmed cases. For instance, despite identical implementation dates, Scenario 3 (5-day isolation) resulted in a much higher cumulative incidence than Scenario 2 (7-day isolation implemented on 10 February 2022).

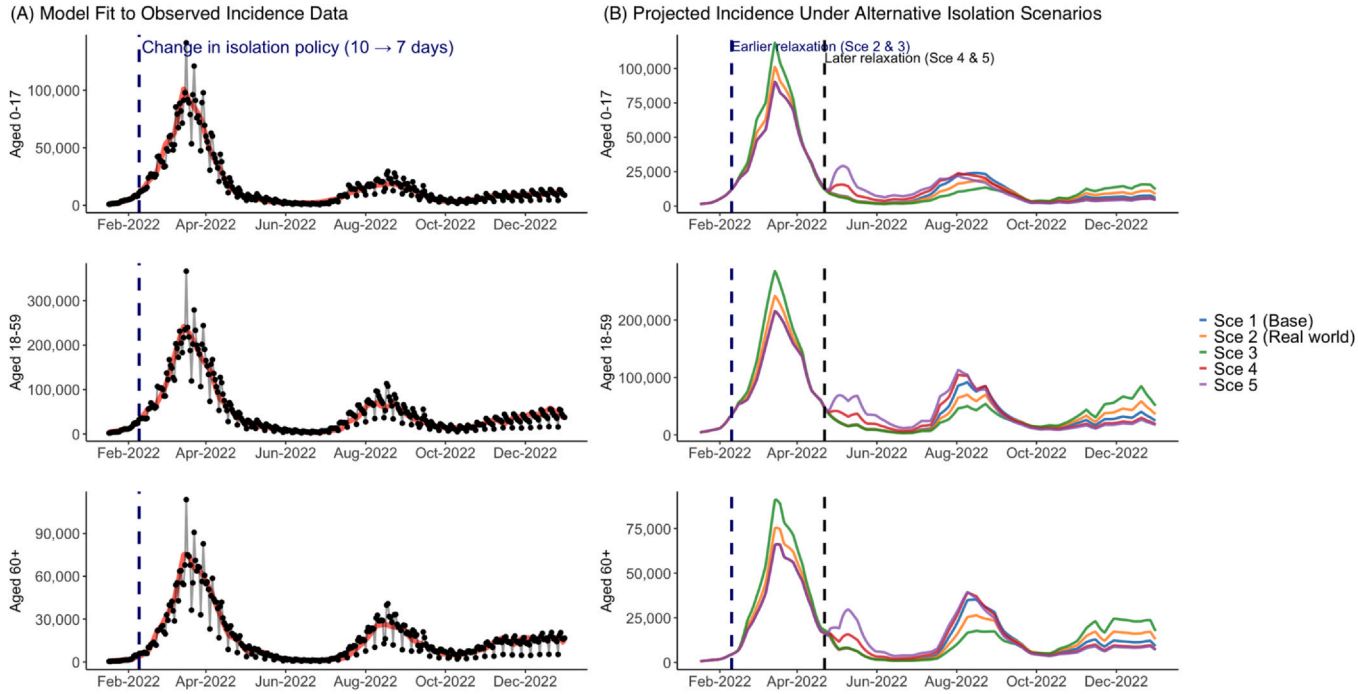

Fig. 2. Model fit and projected epidemic trajectories under alternative isolation period and timing scenarios. Panel (A) shows the model fit to the observed data by age group during 2022. Panel (B) illustrates projected incidence under each scenario. Scenario 1 maintains a 10-day isolation period, while Scenarios 2 and 3 reduce the duration to 7 and 5 days, respectively, starting in February (reflecting the actual policy change). Scenarios 4 and 5 apply the same reductions beginning in April, when population immunity was highest.

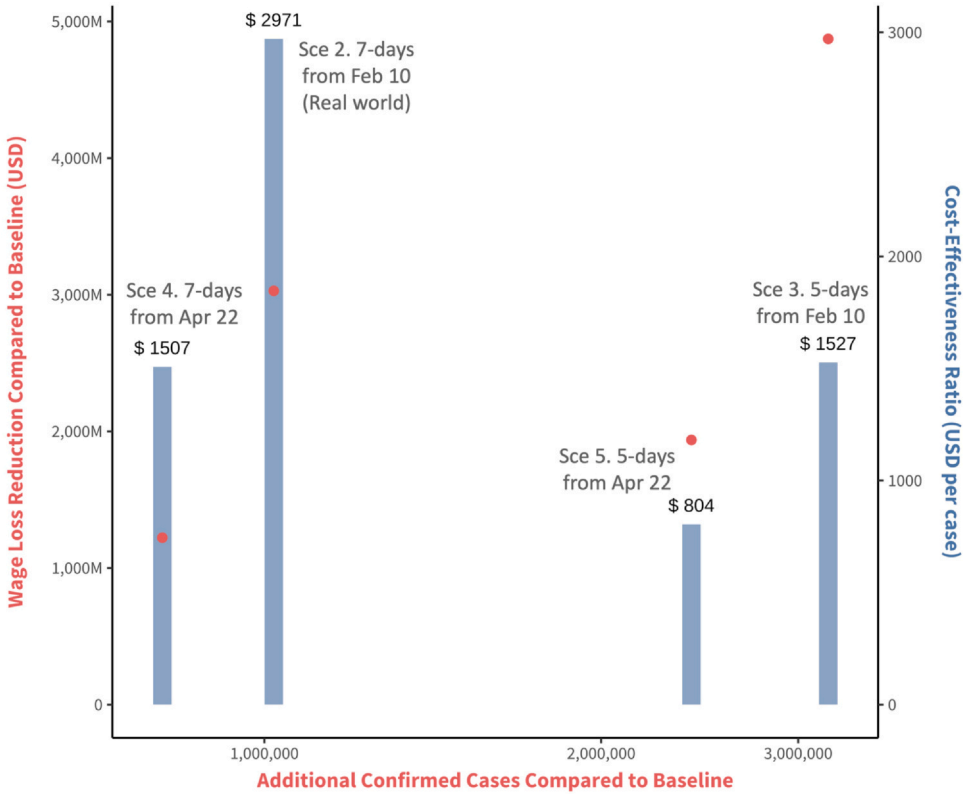

Fig. 3. Cost-effectiveness of isolation period and timing scenarios. Incremental cost-effectiveness of reducing the isolation period and altering the timing of implementation, expressed as total wage loss reduction (USD) per additional COVID-19 case. Bars represent modified cost-effectiveness ratios (CERs), while points indicate the absolute reduction in total wage losses compared to the baseline (10-day isolation maintained). The horizontal position of each point reflects the number of additional cases relative to the baseline.

Fig. 3 presents the trade-off between increased infections and reduced wage losses. Compared to the baseline (Scenario 1: maintaining the 10-day isolation period), all alternative scenarios produced economic savings, albeit at the cost of increased transmission. The modified cost-effectiveness ratios, defined as the reduction in wage loss per additional confirmed case, varied across scenarios. Scenario 2, which reflects the real-world implementation of a 7-day isolation starting on 10 February, exhibited the highest ratio among the evaluated scenarios, reflecting a relatively larger wage loss reduction per additional confirmed case. In contrast, more aggressive isolation shortening (e.g., 5-day isolation starting in February) led to the greatest wage loss reduction, but also to a significant increase in cumulative case numbers. Detailed numerical results are provided in Supplementary Material 4.

Impact of the timing of isolation policy lifting (Period 2: January–August 2023)

Fig. 4 presents the simulation results for Period 2 (January to August 2023), which encompasses the period during which mandatory isolation policies were lifted. Panel (A) shows the model fit to the observed incidence data, capturing the timing and relative magnitude of epidemic peaks across age groups. Panel (B) shows the projected epidemic trajectories under alternative scenarios for the timing of the policy change. Across all age groups, earlier lifting consistently resulted in earlier epidemic peaks, higher incidence levels, and longer epidemic durations, demonstrating the importance of policy timing in shaping transmission dynamics. These patterns were observed within the study period and reflect short-term transmission dynamics following policy changes. The highest cumulative incidence over the entire study period was observed in Scenario 1 (lifting on 1 February 2023), while Scenario 7 (lifting on 1 July 2023) resulted in the lowest cumulative incidence.

When restricting the comparison to the 60-day period following each policy implementation, Scenario 4 (lifting on 1 May 2023) yielded the highest number of cases, whereas Scenario 2 (lifting on 1 March 2023) consistently showed the lowest burden across both the entire period and the 60-day window.

Fig. 5 presents summary comparison of trade-offs for each scenario, measured as the reduction in total wage loss per additional confirmed case. Using the baseline scenario (Scenario 7: lifting on 1 August 2023) as a reference, both additional case counts and wage loss reductions tended to increase with earlier policy implementation. However, the resulting modified cost-effectiveness ratios did not follow a monotonic trend with respect to timing. Scenario 6, implemented 1 month after the real-world policy, exhibited the highest ratio among the alternative scenarios. This reflects differences in the relative magnitude of wage loss reduction per additional case across policy timings. Detailed numerical results are provided in Supplementary Material 5.

Discussion

This study evaluated the epidemiological and economic outcomes of relaxing COVID-19 isolation policies in South Korea using an age-stratified compartmental model calibrated to national surveillance data. Counterfactual scenarios were simulated by varying both the timing and duration of isolation to assess the trade-offs between transmission control and wage-related productivity loss. The study period was divided into two distinct phases: partial relaxation in 2022 (Period 1) and complete lifting of mandates in 2023 (Period 2). Policies that were strategically timed, such as a 7-day isolation policy implemented before the epidemic peak or the full lifting of mandates during later stages of the epidemic, were associated with more favorable trade-off profiles when jointly considering infection control and economic burden. These findings are

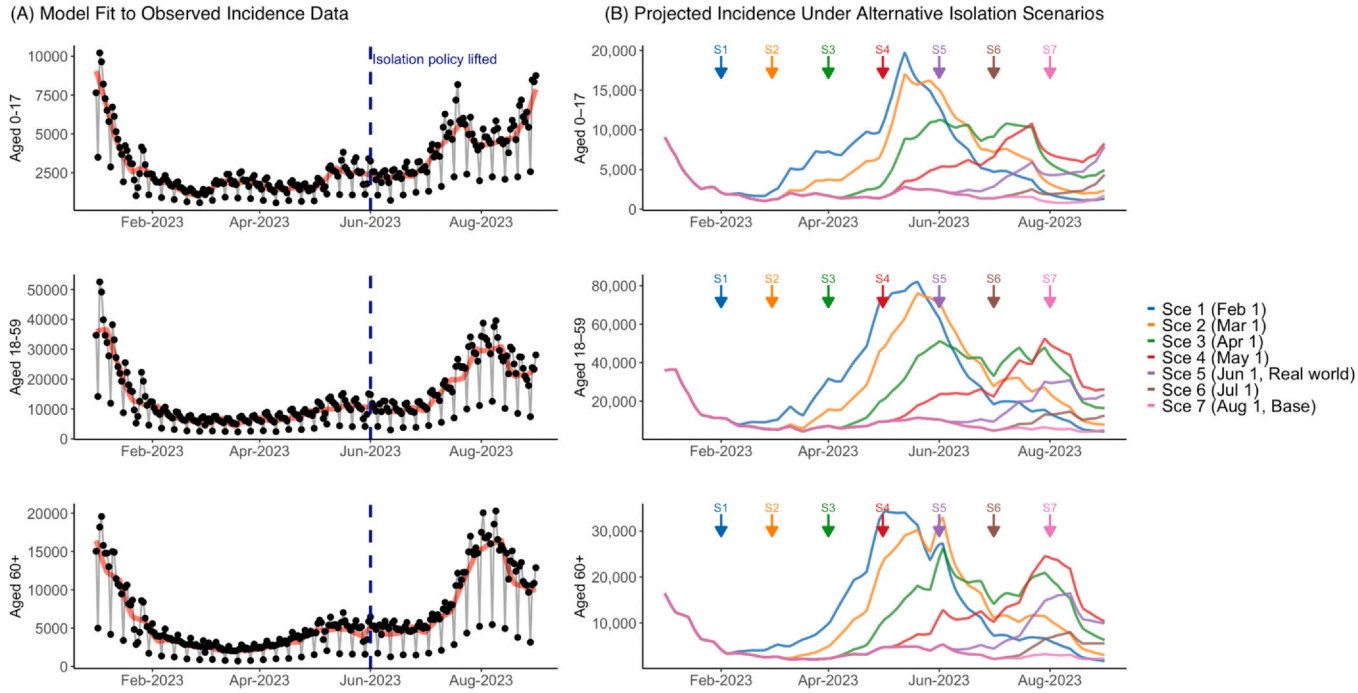

Fig. 4. Model fit and projected epidemic trajectories by timing of isolation policy lifting. Model calibration and projections from January to August 2023, covering the period during which mandatory isolation policies were fully lifted. Panel (A) displays model fits to age-stratified observed incidence data. Panel (B) presents projected epidemic trajectories under different policy lift dates, with arrows indicating the timing of isolation lifting for each scenario. The baseline scenario assumes isolation was lifted on 1 August 2023.

interpreted within the epidemiological and policy context of the Omicron-dominant period, during which South Korea had shifted from suppression-oriented strategies to mitigation-focused adjustments.

In the case of partial relaxation, our results indicated that the extent of isolation reduction had a greater impact on outcomes than the timing of implementation. A 5-day isolation period, compared to a 7-day policy implemented at the same time (e.g., in February), resulted in significantly higher cumulative incidence with only marginal economic gains, yielding a less favorable marginal trade-off profile. This more aggressive approach also produced the highest initial epidemic peaks and subsequent rebounds across all age groups. In contrast, a 7-day isolation policy introduced before the epidemic peak mitigated these surges and was associated with a

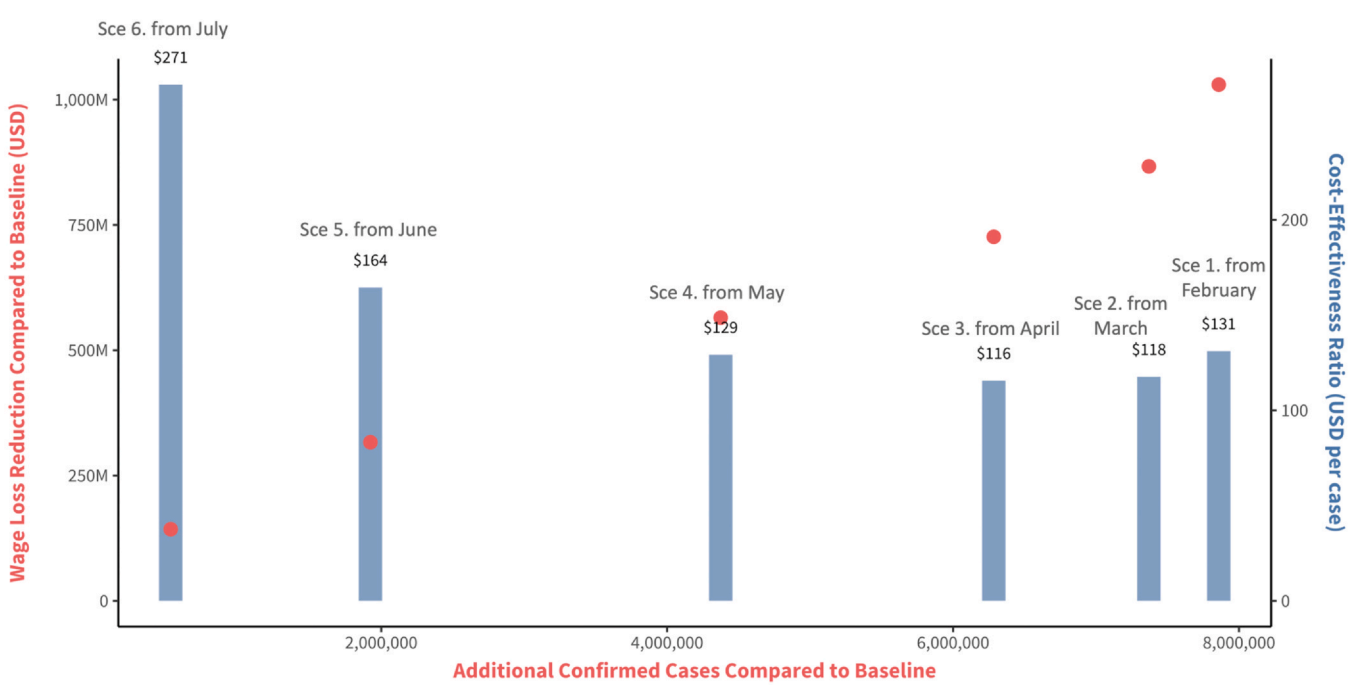

Fig. 5. Cost-effectiveness of isolation policy timing scenarios. Relationship between additional confirmed cases and wage loss reduction relative to the baseline scenario (lifting isolation on 1 July 2023). Each point represents a scenario; bars indicate scaled CERs. Earlier lifting is associated with higher incidence, but also greater economic benefit, reflecting policy trade-offs between health burden and economic impact.

more balanced trade-off between health and economic outcomes. Introducing the same policy later (e.g., in April) achieved comparable reductions in incidence and wage loss, but yield smaller marginal economic gains per additional case, demonstrating the importance of timing even in moderate policy scenarios.

In scenarios involving the complete lifting of isolation mandates, outcomes varied according to the timing of implementation. Early lifting (February–April) was associated with reduced wage loss but resulted in earlier and larger epidemic peaks within the observation period, leading to relatively less favorable trade-off profiles. In contrast, lifting isolation in July, one month after the real-world policy, was followed by delayed and milder epidemic peaks during the study period. These differences may reflect evolving epidemic conditions, as later policy lifting coincided with substantially altered transmission dynamics shaped by widespread infections during the early Omicron surges. Consistent with this interpretation, previous modeling studies have shown that delaying the full lifting of interventions may reduce the risk of short-term rebounds, when higher levels of population immunity have accumulated [15]. From an economic perspective, these later-lifting scenarios were associated with more balanced trade-offs between infection control and wage loss, as reflected in the relative magnitude of wage loss reduction per additional confirmed case.

To our knowledge, few studies have specifically isolated the epidemiological and economic effects of modifying isolation durations as a standalone policy, separate from broader non-pharmaceutical interventions. Most existing models evaluate isolation in combination with other measures, limiting the ability to understand its independent contribution [31]. This gap restricts policymakers' ability to make calibrated decisions about isolation under dynamic epidemic conditions. By simulating isolation-specific scenarios and explicitly quantifying the trade-offs between public health and economic burden, our study addresses this critical gap. This approach enables decision-makers to assess not only whether to lift or relax isolation, but also when and to what extent, based on the epidemic trajectory and prevailing resource constraints.

Prior studies have shown that the economic burden of COVID-19 extends beyond direct medical expenditures, with indirect costs such as productivity loss constituting a substantial component of total societal impact. Multi-country economic burden studies have reported that indirect costs constitute a large share of total COVID-19-related costs, in some settings exceeding direct healthcare expenditures. At the macroeconomic level, COVID-19-related losses have been reported to exceed 10 % of GDP globally and over 20 % in some high-income countries [32]. However, economic evaluations of COVID-19 interventions remain heterogeneous in analytic perspective and cost components, and often incompletely capture non-health impacts associated with non-pharmaceutical interventions [21]. Accordingly, this study focuses on wage loss attributable to isolation as a measurable and policy-relevant component of the economic burden, enabling internally consistent comparisons across alternative isolation scenarios within a single national context. Consistent with this perspective, prior population-based analyses have shown that productivity loss attributable to COVID-19 can be substantial even over relatively short time horizons—for example, with estimated losses of approximately EUR 143 million among working-age adults, corresponding to around 0.08 % of national GDP, alongside additional absenteeism-related losses [30]. Moreover, simulation-based economic evaluations have demonstrated that policy conclusions can differ materially once such economic impacts are incorporated, with strategies that appear favourable when assessed using health outcomes alone generating negative net monetary value after accounting for broader economic losses associated with physical distancing or isolation [18].

Building on this interpretation, we suggest that tailored policy adjustments, such as moderate relaxation prior to epidemic peaks and full lifting once population immunity is sufficiently high, can help optimize the balance between public health and economic burden. Our findings highlight the importance of adaptive response to real-time epidemic conditions. If such evaluations can be conducted proactively, rather than only retrospectively, public health authorities could use model-based projections to identify the optimal timing and intensity of policy shifts. To support these adaptive decisions, a robust scientific infrastructure, including real-time surveillance, immunity monitoring, and predictive modeling, is essential [33]. These tools not only facilitate epidemic forecasting, but also provide comparative insights across alternative scenarios, enabling policymakers to assess both health and economic impacts within an integrated framework. The importance of investing in such infrastructure has also been emphasized in recent studies on pandemic decision-making and evidence-informed policy [33].

This study offers several methodological and policy contributions. First, it presents a scenario-based modeling framework that jointly considers infection control and socioeconomic burden under varying intervention strategies. Second, rather than aiming for precise predictions amid substantial epidemiological uncertainty, this study emphasizes relative comparisons across plausible scenarios, providing a practical approach for selecting cost-effective policies when real-time data are limited.

However, several limitations should be acknowledged. First, wage loss was used as a proxy for economic burden of COVID-19 policies. While this measure captures a policy-relevant component of societal cost, it limits direct quantitative comparison with other economic evaluations that apply different economic frameworks. In addition, the modified cost-effectiveness ratio was applied to summarize marginal trade-offs relative to a conservative reference scenario. Accordingly, this indicator should be interpreted as a comparative measure of incremental economic gains associated with policy relaxation, rather than as a criterion for selecting an overall optimal strategy across all policy alternatives. Second, the model did not differentiate between subvariants of Omicron, which may exhibit varying transmission and clinical characteristics. Such differences could affect the absolute magnitude of infection and wage-loss estimates, although relative comparisons across policy scenarios are likely to remain robust. Finally, indirect behavioral changes, such as altered contact patterns following policy adjustments, were not explicitly modeled and were assumed to remain consistent across all scenarios. Incorporating these dynamics could allow the model to more closely reflect real-world transmission dynamics and represents an important direction for future research.

Future directions

Although this study did not aim to predict exact infection numbers, it provides a comparative, scenario-based framework for evaluating policy options under dynamic epidemic conditions. Building on this foundation, future research should focus on reducing model uncertainty and more accurately reflecting real-world behavioral dynamics to improve the applicability of such policy evaluation framework. In particular, the integration of data-driven approaches, such as machine learning for improving epidemic forecasting, capturing real-world behavioral dynamics, and refining parameter estimation, may enhance model adaptability and improve the timeliness and precision of scenario-based projections. Strengthening real-time surveillance systems and immunity monitoring, alongside advances in predictive modeling, will be essential to operationalize such scenario-based evaluations of control policies and to support proactive and adaptive decision-making during future epidemics.

Conclusions

By quantifying the trade-offs between infection control and economic burden under alternative isolation policy scenarios, this modeling study offers a practical framework to support policy

decision-making during future epidemics. Our findings highlight the importance of evaluating both the timing and extent of policy changes in advance, supporting more timely and sustainable responses. To operationalize such scenario-based approaches in real time, investments in scientific infrastructure, including surveillance systems, immunity monitoring, and predictive modeling, will be essential. These efforts will help strengthen pandemic preparedness and facilitate context-specific, data-driven policy adjustments, particularly in the diverse national and regional settings.

Author's contributions

Conceptualization: SJ and KM. Data curation: SJ and KM. Formal analysis: SJ. Methodology: SJ and KM. Project administration and funding acquisition: SC and AC. Supervision: KM. Writing—original draft: SJ and KM. Writing—review and editing: SJ, SC, AC, and KM. All authors had full access to the data, reviewed and edited the manuscript, and accept responsibility for submitting the paper for publication.

Ethical approval

This study used aggregated data that are publicly accessible. No ethical approval was required.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) to assist with English language editing. After using this tool, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jiph.2026.103160.

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